

EMC ASSESSMENT

Product: SidecarTridge TOS Emulator Mega ST Kit

Model/Revision: SidecarTridge TOS Emulator Mega ST Kit v4 (Rev.4.0.0) — TOS Emulator module + 256KB Decoder Board / REV4.0.0

1. PRODUCT OVERVIEW

The product is a low-voltage digital electronic device intended for use in retro computing systems. It is designed as an internal or semi-internal component (e.g. expansion board, cartridge, or interface module) and is powered by the host system.

Typical operation voltage is below 50 V DC (e.g. 3.3V / 5V / 12V).

2. APPLICABLE DIRECTIVE

This assessment is made in relation to Directive 2014/30/EU (Electromagnetic Compatibility - EMC).

The product does not fall under Directive 2014/35/EU (Low Voltage Directive) as it operates below 50 V DC.

3. DESIGN CHARACTERISTICS RELEVANT TO EMC

- The device contains standard digital logic components (e.g. CMOS/TTL, microcontrollers, memory devices).
 - No radio transmitter or intentional RF radiator is present.
 - The device does not include wireless communication functionality.
 - The design does not incorporate high-frequency switching power supplies (unless provided externally by the host system).
 - Signal frequencies are limited to those typical of legacy computing platforms (generally low MHz range or below).
 - External connections are limited in length and are typically internal to the host system or via short cables.
 - The device includes a microcontroller-based system operating at moderate clock frequencies typical of embedded systems.
 - A USB interface is present for local data transfer; it is not intended for continuous high-speed communication during normal operation.
 - The product consists of two internal sub-assemblies (a microcontroller-based TOS Emulator module and a passive address-decoder board) operating together as a single system.
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4. EMISSION ASSESSMENT

Due to the design characteristics described above:

- The product is not expected to generate significant electromagnetic emissions.
- Any emissions are comparable to those of standard digital logic circuits used in legacy computing systems.
- The absence of high-frequency switching regulators and RF transmitters significantly reduces emission risk.
- The USB interface is used intermittently for file transfer and does not represent a continuous high-frequency emission source.

Conclusion:

The product presents a low risk of causing electromagnetic interference when used as intended.

5. IMMUNITY ASSESSMENT

- The product is designed to operate within the electromagnetic environment provided by the host system.
- Immunity performance depends on the overall system in which the product is installed.
- The product does not include specific shielding or filtering beyond standard design practices.

Conclusion:

The product is expected to operate correctly in typical residential, commercial, and light-industrial environments when integrated into a compliant host system.

6. INSTALLATION AND USE CONDITIONS

- The product is intended for integration into a larger electronic system.
 - It is not designed to operate as a standalone end-user device.
 - Proper operation and EMC performance depend on correct installation within a compliant host system.
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7. OVERALL CONCLUSION

Based on the design characteristics and intended use, the product is considered to comply with the essential requirements of Directive 2014/30/EU (EMC), provided that it is installed and used in accordance with its intended purpose.

This assessment is supported by the use of harmonised standards such as:

- EN IEC 61000-6-1:2019 (Immunity)
- EN IEC 61000-6-3:2021 (Emissions)

No additional EMC mitigation measures are required based on the design and intended use.

8. RESPONSIBLE PERSON

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Date: 23 April 2026

Signature:

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